LoR-SGS: Hyperspectral Image Compression via Low-Rank Spectral Gaussian Splatting

Tianyu Li, Ting Wang, Xile Zhao, *Senior Member, IEEE*, and Chao Wang

***Abstract*—Hyperspectral images (HSIs), composed of hundreds of spectral bands per pixel, require efficient compression methods to achieve compact representations while preserving reconstruction quality. Implicit neural representations (INRs) have shown promise but struggle to efficiently model the spatial locality of HSIs, resulting in high GPU memory usage and long encoding times. To mitigate these challenges, we propose LoR-SGS, a novel framework that leverages Gaussian splatting (GS) for HSI compression. GS adaptively represents images as a collection of Gaussians, parameterized by position, covariance, and color attributes, enabling efficient rendering and compact storage. However, direct application to HSIs is hindered by the prohibitive spectral dimensionality, and a naive extension of GS for RGB images would severely distort the high-dimensional spectral signatures, leading to substantial degradation in reconstruction fidelity. In response, we introduce spectral GS (SGS) under a low-rank decomposition. LoR-SGS mitigates the parameter explosion caused by high spectral dimensions while efficiently capturing spatial details. Additionally, the coefficient basis matrix is estimated using nonnegative matrix factorization (NMF). All Gaussian attributes and the coefficient matrix are further quantized and entropy-coded into a practical codec. Evaluations on four benchmark datasets demonstrate that LoR-SGS achieves significant performance improvements, delivering on average 26% higher peak signal-to-noise ratio (PSNR) than state-of-the-art INR-based compressors at 0.1 bits-per-pixel-per-band (bpppb), while reducing GPU memory usage by over 10% and decreasing encoding time by over 20%. Code is available at https://github.com/JackTianyuLi/LoR-SGS**

***Index Terms*—Gaussian splatting (GS), hyperspectral image (HSI) compression, low rank decomposition.**

I. Introduction

HYPERSPECTRAL images (HSIs) record hundreds of contiguous spectral bands per pixel, facilitating fine-grained material classification and environmental analysis in fields like remote sensing [1], agriculture [2], and medical diagnostics [3]. However, the HSI data cube's high dimensionality across spatial and spectral axes creates heavy burdens on onboard storage, bandwidth-limited transmission, and latency-sensitive real-time processing [4], [5]. To overcome these challenges, efficient HSI compression is essential to alleviate these constraints without degrading the spatial and spectral fidelity for downstream applications. Various approaches to HSI compression have been proposed, which can generally be classified into three categories: conventional methods, decomposition-based methods, and deep learning-based models.

Conventional codecs (e.g., JPEG [6], JPEG2000 [7]) leverage block-based spatial transforms but ignore rich inter-band correlations. Although suitable for the RGB imagery, these approaches suffer from blocking artifacts and distort narrow band spectral features, undermining material discrimination. To exploit inter-band dependencies, hybrid methods such as principal component analysis (PCA)-DCT [8], PCA-JPEG2000 [7], and 3D-DCT [9] apply PCA for spectral decorrelation prior to spatial coding. Nonetheless, they retain blocking artifacts and lack the flexibility to model nonlinear spectral mixtures at material boundaries. In addition, video-codec extensions (HEVC [10], H.264/AVC [11]) view bands as frames and fail to capture complex spectral dynamics only via the simplistic inter-frame predictors. Therefore, compression methods in RGB images or video data cannot simply adapt to the HSI problems owing to their high-dimensionality and spectral properties.

Decomposition-based methods leverage the low-rank or multilinear structure of HSIs by factorizing the data cube into spatial and spectral bases for compact coding [12], [13]. Patch-based low-rank tensor decomposition (PLTD) clusters similar local patches into a fourth-order tensor and applies Tucker/SVD factorization to each cluster [14]. Block-sparse tensor methods extend this idea by learning overcomplete dictionaries that sparsely encode third-order patches, but they incur high dictionary-learning and reconstruction overheads, especially at low bitrates [15]. Sparse Tucker decompositions incorporate sparsity priors to reduce core-tensor size, yet they still struggle to capture nonlinear spectral transitions at material boundaries, leading to degradation in reconstruction fidelity [16]. Despite attempts to manage the high dimensionality of HSI data, decomposition-based methods often struggle with challenges such as high computational costs, poor compression ratios, and lossy reconstruction that introduces artifacts and compromises fine spectral details

Received 5 July 2025; revised 9 September 2025; accepted 12 October 2025. Date of publication 20 October 2025; date of current version 30 October 2025. This work was supported in part by the Natural Science Foundation of China under Grant 12201286, Grant 12371456, Grant 12171072, and Grant 62131005; in part by Guangdong Basic and Applied Research Foundation under Grant 2024A1515012347; in part by Shenzhen Science and Technology Program under Grant 20231115165836001; in part by Sichuan Science and Technology Program under Grant 2024NSFJQ0038 and Grant 2024NSFSC0038; and in part by the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation under Grant pdjh2025c20501. *(Corresponding author: Chao Wang.)*

Tianyu Li, Ting Wang, and Chao Wang are with the Department of Statistics and Data Science, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: 12212824@mail.sustech.edu.cn; 12231293@mail.sustech.edu.cn; wangc6@sustech.edu.cn).

Xile Zhao is with the School of Mathematical Science, University of Electronic Science and Technology of China, Chengdu 610054, China (e-mail: xlzhao122003@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3623253

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

critical for analysis. These issues emphasize the need for a more compact, near-lossless representation that captures and preserves intricate spatial and spectral details without requiring explicit pixel-wise storage.

With the rapid advancement of deep learning, learning-based compression schemes have gained increasing prominence. Recent developments have introduced hybrid codecs that combine traditional transformation techniques with deep neural networks, effectively leveraging learned spatial–spectral priors for improved performance. Among these, line-based attention predictors [17] have emerged as a powerful approach, performing recursive spectral prediction on a line-by-line basis prior to entropy coding. GAN-based approaches incorporate spatial-spectral priors into HiFiC-style architectures, using 3D convolutions or squeeze-excitation blocks to enhance texture fidelity at low bitrates [18]. Multisensor autoencoders utilize a two-stage training strategy with spatial and spectral training to generate compact latent codes that achieve competitive peak signal-to-noise ratio (PSNR), though requiring careful balancing between the two phases [19]. Edge-guided dual-attention networks prealign patches along material boundaries, reducing boundary artifacts while incurring side-information overhead [20]. Contrastive-learning-based compressors improve channel-wise embeddings by maximizing inter-band separability, but they depend on large contrastive batches and complex training schedules [21]. Other state-of-the-art learning-based methods employ implicit neural representations (INRs), which enables a continuous and efficient representation of HSI data, preserving fine-grained spatial and spectral details [22], [23]. Despite notable improvements in compression efficiency and reconstruction fidelity. However, INRs typically use a high-dimensional multilayer perceptron (MLP) network to effectively represent HSIs, which demands thousands of optimization steps and significantly increases GPU memory usage. As a result, encoding times are prolonged, limiting the feasibility of on-device or real-time applications. Additionally, the MLP parameters serve as the learned representation of the input signal, making it difficult to achieve an optimal balance between quality and compression efficiency.

To mitigate the computational and memory challenges of deep learning models and the limitations of decomposition-based methods, we propose a novel HSI compression technique that enables efficient training, optimized GPU memory usage, and faster encoding. Our approach leverages Gaussian splatting (GS) [24], a recent advancement in 3-D scene reconstruction. GS adaptively represents 3-D scenes as a collection of 3-D Gaussians, parameterized by position, covariance, and color attributes, enabling efficient rendering and compact storage. Leveraging its inherent advantages, including fast rendering speed, short training time, low GPU memory cost, and spatial continuity, GS has recently been extended to 2-D scenarios, demonstrating remarkable performance in color image representation and compression, particularly in retaining details [25]. Nevertheless, it is nontrivial to directly adapt 3DGS or 2DGS to the HSI compression. First, existing 3DGS methods depend on varying camera transformation matrices to render multiview images, which involves a vast number of learnable parameters. This naive implementation imposes high storage and communication overhead for HSI representation. Second, since HSIs contain numerous spectral bands, a straightforward adaptation of 2DGS designed for RGB images is clearly impractical. If each band's image were represented using 2DGS, it would also introduce a substantial number of parameters, further increasing the storage and communication requirements for HSI representation.

To address these challenges, we propose a novel HSI representation and compression framework. First, we exploit the inherent low-rank structure of HSIs by decomposing the HSI matrix into the reduced spectral image and coefficient matrices (or endmembers and abundances) via matrix factorization. Second, since the reduced spectral image is significantly smaller than the original HSI data, we propose a spectral GS (SGS) to represent it via a weighted mixture of 2-D Gaussians. Finally, we transform this Gaussian-based representation into a practical image codec. Our key contributions are as follows.

1) We propose a novel representation and compression scheme, LoR-SGS, for HSIs, leveraging GS within a low-rank matrix decomposition framework. Additionally, an optimization model is designed to directly update Gaussian parameters through low-rank decomposition, eliminating cumulative errors from intermediate outputs.
2) For the first time, we propose SGS to model reduced spectral images using compact 2-D Gaussian representations and an efficient rasterization scheme. This approach achieves superior representation quality with fast compression while preserving spatially adaptive modeling through GS's flexible representation power.
3) Extensive experiments on four benchmark HSI datasets demonstrate that the proposed LoR-SGS method achieves state-of-the-art compression performance, delivering on average 26% higher PSNR than state-of-the-art INR-based compressors at 0.1 bits-per-pixel-per-band (bpppb) across diverse scenes.

The remainder of this article is organized as follows. Section II introduces the related work with GS and decomposition-based methods of HSI Compression. Section III shows the formulation of 3DGS as preliminaries. Section IV presents the proposed model and its implementation details. Experimental results and ablation discussion are presented in Section V. Finally, Section VI draws a conclusion and future work.

II. Related Work

A. Decomposition-Based HSI Compression

Decomposition-based approaches capitalize on the intrinsic low-rank structure of HSIs, where the entire 3-D image cube can be effectively represented through combinations of fundamental spectral signatures and their spatial distributions. A typical strategy involves decomposing the HSI into a set of constituent spectral signatures (endmembers) and their associated spatial abundance distributions, with both components constrained to be nonnegative [14], [26]. The core principle lies in approximating the original HSI through a limited collection of representative spectral bases and their per-pixel mixing coefficients. These bases capture distinct material signatures, while the coefficients indicate their spatial presence across the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

scene. This decomposition can be achieved through various techniques, ranging from random assignments (for iterative methods) to more sophisticated spectral analysis approaches like singular value decomposition (SVD) and vertex component analysis (VCA). Once derived, the compact representation enables efficient compression, particularly through additional encoding of the spatial abundance patterns where redundancy can be effectively exploited [4]. However, traditional encoding of these decomposed components (e.g., through simple quantization or standard image codecs) often leads to significant reconstruction errors and limits the achievable fidelity, hindering the "near-lossless" potential of such methods. In our work, we incorporate this dual-stage approach (decomposition + encoding) by further compressing the resulting reduced spectral via SGS. This method offers a more flexible and expressive continuous representation that leads to higher fidelity reconstruction at a given bitrate compared to previous decomposition-based methods.

B. GS and Its Recent Applications

3-D GS(3DGS) [24] models a scene by a set of anisotropic Gaussians that can be rasterized efficiently with α-blending, with each having distinct parameters for center, covariance, and color. 3DGS affords fast rasterization and compact memory use, outpacing mesh or point-cloud approaches on dynamic scenes and fine geometry [24]. Survey work [27] reviews 3DGS for real-time implicit surface rendering, point-cloud upsampling, dynamic scene interpolation, and relighting, demonstrating broad applicability. There has also been a surge in various application domains where 3DGS has been proven valuable, such as 3-D scene compression [28] and editing [29].

Recent work extends 3DGS to 2-D image compression by modeling RGB images or even videos as mixtures of 2-D Gaussians, where each Gaussian encodes local color and spatial features. This trend has rapidly expanded. For instance, GaussianImage achieves INR-level quality at over 1000 FPS with minimal memory requirements [25]. Building on this, extensions like MiraGe enable real-time editing [30], while hierarchical splatting scales seamlessly to gigapixel panoramas without introducing tiling artifacts [31]. In video representation and compression, methods such as D2GV and GaussianVideo deliver artifact-free decoding at hundreds of FPS while significantly reducing bitrates compared to HEVC [32], [33]. These innovations highlight the growing potential of GS-based techniques across diverse applications. Despite these advances, there has been little exploration of using GS on HSIs. Current 2DGS methods neither account for the intrinsic spectral structure of HSIs nor are designed to handle hundreds of spectral bands. Furthermore, directly extending 2DGS [25], to HSI would require modeling each spectral band independently, multiplying parameter counts by the number of bands. This negates compression gains and ignores HSI's low-rank spectral correlations.

III. Preliminaries

3DGS is a recent inverse rendering paradigm that represents a 3D scene using a collection of anisotropic Gaussian primitives, each defined by a multivariate Gaussian function [24]. These primitives serve as continuous volumetric elements that can be directly rendered to screen space using differentiable splatting operations.

Formally, each Gaussian in 3-D position $\mathbf{x} \in \mathbb{R}^3$ is defined as

$$G(\mathbf{x}) = \exp\left(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right) \tag{1}$$

where $\boldsymbol{\mu} \in \mathbb{R}^3$ denotes the center position and $\boldsymbol{\Sigma} \in \mathbb{S}^3_{++}$ is a symmetric positive-definite covariance matrix controlling the shape and orientation of the Gaussian ellipsoid in 3-D space. Note that this formulation omits the normalizing constant, as our Gaussians serve as basis functions rather than probability density functions. To enable projection and rendering, the 3-D Gaussian is transformed into 2-D image space through a differentiable pipeline. The covariance matrix $\hat{\boldsymbol{\Sigma}}$ of the projected 2-D Gaussian is computed as

$$\hat{\boldsymbol{\Sigma}} = \mathbf{JW}\boldsymbol{\Sigma}\mathbf{W}^\top\mathbf{J}^\top \tag{2}$$

where $\mathbf{W}$ is the local view transformation matrix, and $\mathbf{J}$ is the Jacobian matrix of the perspective projection at the current viewpoint [24], [34]. The final pixel color $\mathbf{c}$ is computed through α-blending from front to back

$$\mathbf{c} = \sum_{i=1}^{N} \mathbf{c}^{(i)}\alpha^{(i)}\prod_{j=1}^{i-1}\left(1-\alpha^{(j)}\right) \tag{3}$$

where $\mathbf{c}^{(i)} \in \mathbb{R}^3$ and $\alpha^{(i)} \in \mathbb{R}$ denote the color and opacity of the ith projected Gaussian, respectively. The opacity $\alpha^{(i)}$ is obtained based on $\hat{\boldsymbol{\Sigma}}$

$$\begin{aligned}\alpha^{(i)} &= o^{(i)}\exp\left(-\sigma^{(i)}\right)\\ \sigma^{(i)} &= \tfrac{1}{2}\left(\mathbf{x}-\boldsymbol{\mu}^{(i)}\right)^\top\left(\hat{\boldsymbol{\Sigma}}^{(i)}\right)^{-1}\left(\mathbf{x}-\boldsymbol{\mu}^{(i)}\right)\end{aligned} \tag{4}$$

where $\exp(-\sigma^{(i)})$ is the projected density and $o^{(i)}$ the view-dependent visibility. To ensure that the 3-D covariance matrix $\boldsymbol{\Sigma}$ remains positive semi-definite, it is parameterized via a rotation-scaling decomposition $\boldsymbol{\Sigma} = \mathbf{RSS}^\top\mathbf{R}^\top$, with rotation matrix $\mathbf{R}$ and diagonal scaling matrix $\mathbf{S} = \mathrm{diag}(s_x, s_y, s_z)$. 3DGS enables efficient volumetric rendering, owing to its differentiability and compactness.

For a typical 3-D scene, 3DGS represents it as a collection of 3-D Gaussians, each defined by a learnable 3-D mean ($\boldsymbol{\mu}$), a 3-D covariance matrix ($\boldsymbol{\Sigma}$), a scalar opacity (o), and a view-dependent color. For a 2-D image, the process is similar. It involves learning a set of 2-D Gaussians, with each having its mean (2-D position), a 2-D covariance matrix, and a spectral vector $\mathbf{c}$. Our method utilizes this approach, with the number of components in $\mathbf{c}$ equal to the number of bands of the image.

IV. Methodology

A. Overall Framework

Let the HSI data cube be $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$, where h, w are spatial dimensions and l is the number of spectral bands. It is worth noticing that HSI data has strong spectral correlation and spectral redundancy, which are modeled as a low-rank property. In order to compress HSI, we first utilize this low-rank property and represent the data as

$$\mathcal{X} \approx \mathcal{A} \times_3 \mathbf{E} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

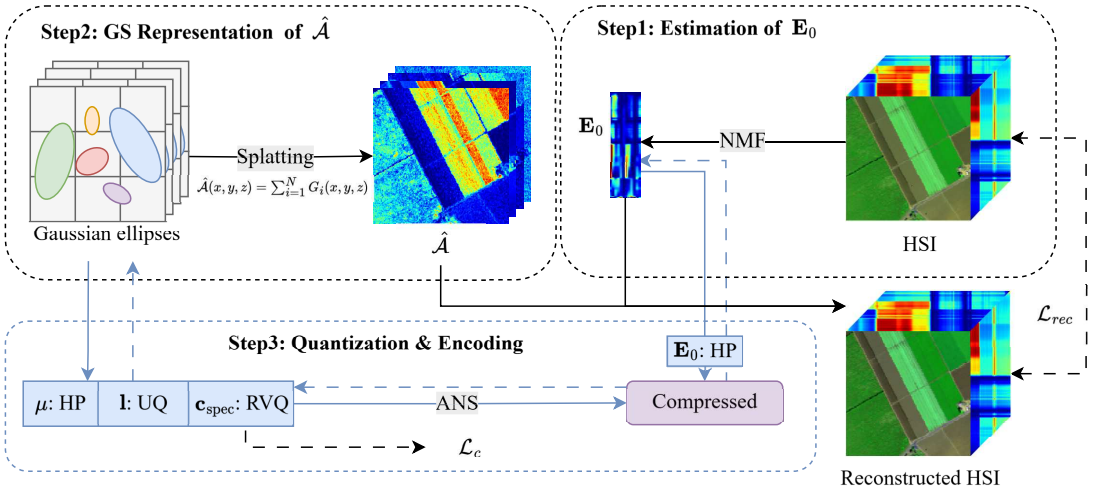


Fig. 1. Overall pipeline of our method. Here, HP, UQ, RVQ, and ANS refer to HP (16-bit), UQ, RVQ, and ANSs, respectively.

where $\times_3$ is mode-3 multiplication between the reduced spectral image $\mathcal{A} \in \mathbb{R}^{h\times w\times k}$ and the coefficient basis matrix $\mathbf{E} \in \mathbb{R}^{k\times l}$ with $k \ll l$. This representation decouples spectral bases from spatial weights, leveraging the low-rank property of the HSI data cube. Note that $\mathbf{E}$ has size only $k \times l$, whose storage overhead is negligible. By contrast, the spectral image $\mathcal{A} \in \mathbb{R}^{h\times w\times k}$, though reduced compared to the data cube, is still large, motivating a compact representation.

To mitigate this, we first estimate the coefficient basis matrix $\mathbf{E}$ from the degraded observation using a closed-form initialization (e.g., SVD). After estimating this coefficient matrix, we then focus on modeling the spectral image $\mathcal{A}$ in a highly compact form. Specifically, we represent $\mathcal{A}$ using a collection of spatially adaptive 2-D Gaussians, where each Gaussian carries spectral features through a linear combination of the coefficient basis matrix. These Gaussians act as a sparse, continuous parameterization of the scene, significantly reducing memory usage. The entire restoration process, therefore, consists of two stages, including coefficient basis matrix estimation from the observed HSI, and spectral image reconstruction via differentiable GS. An overview of the pipeline is shown in Fig. 1. In addition, to achieve more aggressive compression, we introduce quantization strategies for model parameters during training and inference. Next, more details on estimating the coefficient matrix $\mathbf{E}_0$ and the reduced spectral image $\mathcal{A}$ are provided.

B. Coefficient Basis Matrix Estimation

The coefficient basis matrix $\mathbf{E} \in \mathbb{R}^{k\times l}$ plays a crucial role in decoupling the spectral and spatial components of the hyperspectral data. Given a degraded HSI observation $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$, our goal is to estimate a low-rank spectral basis $\mathbf{E}$ that captures the dominant spectral information across the scene. While the spectral mixture interpretation is not the primary focus of our work, the structure of the coefficient matrix is analogous to that of endmembers in traditional unmixing formulations, where nonnegative matrix factorization (NMF) is an efficient tool to extract $\mathbf{E}$ in a purely data-driven manner.

First, the HSI cube $\mathcal{X}$ is reshaped into a 2-D matrix $\mathbf{X} \in \mathbb{R}^{hw\times l}$, where each row corresponds to the spectrum of a single pixel. We solve the following NMF problem:

$$\min_{\mathbf{A},\mathbf{E}\geq 0} \|\mathbf{X}-\mathbf{A}\mathbf{E}\|_F^2 \tag{6}$$

where $\mathbf{A} \in \mathbb{R}^{hw\times k}$ is an unfolded reduced spectral image, $\mathbf{E} \in \mathbb{R}^{k\times l}$ contains the estimated spectral bases, and $\|\cdot\|_F$ is the Frobenius norm. The nonnegativity constraints on both $\mathbf{A}$ and $\mathbf{E}$ ensure physical interpretability and are consistent with the characteristics of hyperspectral data. To solve the optimization problem (6), we utilize a coordinate descent update scheme and obtain the minimizer $\{\mathbf{A}_0, \mathbf{E}_0\}$ [35].

This coefficient estimation step using NMF is efficient and unsupervised, and allows the model to flexibly adapt to diverse spectral distributions in real-world HSI datasets.

C. Spectral GS

Here, we consider compressing the spectral image $\mathcal{A}$ by representing it as a weighted mixture of 2-D Gaussians. With the recent advancements in GS [24], the engineering efficiency and compression of Gaussian functions have significantly improved, making them highly suitable for practical implementation. Therefore, we propose a SGS by using Gaussian functions $G_i(x, y, z)$ as fundamental continuous functions to reconstruct a continuous reduced spectral image $\hat{\mathcal{A}}(x, y, z)$

$$\hat{\mathcal{A}}(x,y,z) = \sum_{i=1}^{N} c_z^{(i)} G_i(x,y;\boldsymbol{\Theta}) \tag{7}$$

where N denotes the number of Gaussian kernels, x and y represent the location in the 2-D space, and z represent the location in the spectral dimension. The value of each Gaussian kernel G at the position (x, y) can be expressed as

$$G(x,y;\boldsymbol{\Theta}) = \exp\left(-\tfrac{1}{2}\mathbf{d}^\top \boldsymbol{\Sigma}^{-1}\mathbf{d}\right) \tag{8}$$

where we skip i for notation simplication. The distance vector $\mathbf{d}$ represents the deviations of x and y from their positions μ_x and μ_y. Here, c_z is the zth entry in the spectral attribute

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

vector $\mathbf{c}_{\text{spec}} = [c_1, \ldots, c_k] \in \mathbb{R}^k$. A critical observation arises from the spatial co-registration of spectral bands: all Gaussian parameters (excluding spectral intensity) can be consistently maintained across co-located (x, y) positions in the hyperspectral cube. This spatial–spectral coherence enables shared geometric properties (position and covariance). In addition, we apply Cholesky decomposition on the covariance matrix $\mathbf{\Sigma}$ of each Gaussian kernel to keep positive semi-definite $\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^\top$. We use a Cholesky vector $\mathbf{l} = [l_1+\delta, l_2, l_3+\delta] \in \mathbb{R}^3$ to represent the lower triangular elements in matrix $\mathbf{L}$, where δ is a small nonzero constant in diagonal for preventing singularity of $\mathbf{\Sigma}$. We employ an initialization where $l_1 = l_2 = l_3$, which produces superior results compared to traditional initialization methods. Here, $\boldsymbol{\mu} = [\mu_x, \mu_y] \in \mathbb{R}^2$ to represent the position. Therefore, we implicitly convert the spectral information of the reduced spectral image into the parameters of a 2-D Gaussian. Each 2-D Gaussian kernel has $k + 5$ parameters that need to be optimized, which include $\mathbf{\Theta} := \{\boldsymbol{\mu}, \mathbf{l}, \mathbf{c}_{\text{spec}}\}$.

To achieve simultaneous high-fidelity reconstruction and efficient compression of HSI cubes, our approach fundamentally differs from conventional GS techniques that employ either explicit representation of complete HSI volumes or the estimated spectral images $\mathbf{A}_0$ in NMF formulation (6). Instead, we introduce an optimization framework to update the Gaussian parameters by minimizing the following reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \left\| \mathcal{X} - \hat{\mathcal{A}} \times_3 \mathbf{E}_0 \right\|_F^2 \tag{9}$$

where $\hat{\mathcal{A}}$ denotes the reconstructed spectral image derived from the Gaussian parameters in (7). This formulation offers two distinct advantages: 1) it circumvents error propagation in NMF and 2) it enables efficient self-supervised representation of the compressed spectral imagery. The latter property fundamentally differentiates our method from traditional 3-D or 2DGS implementations. Note that this updating scheme not only can avoid accumulating errors from NMF but also works efficiently to present the reduced spectral image in a self-supervised manner, which distinguishes it from 3-D or 2DGS. This loss brings the optimal Gaussian representation of the HSIs.

D. Quantization and Encoding

To further compress the Gaussian representation, we adopt a quantization and encoding strategy inspired by the methods introduced in GaussianImage [25]. Specifically, we discretize these parameters and then encode the quantized values using efficient coding techniques.

1) Quantization on Parameters and the Coefficient Matrix: Given a set of 2-D Gaussian points to fit the image, we apply distinct quantization schemes to each attribute to mitigate the storage cost while preserving reconstruction quality. In our framework, the 2-D Gaussian is defined by position, covariance (represented via a Cholesky vector $\mathbf{l} = \{l_1, l_2, l_3\}$), and spectral attribute vector. Because the Gaussian positions are sensitive to quantization errors, and the size of $\mathbf{E}_0$ is already small, we use the half-precision (HP) for $\boldsymbol{\mu}$ and $\mathbf{E}_0$. For the covariance parameters $\mathbf{l}$, we adopt a 16-bit asymmetric uniform quantization (UQ) strategy [36] with the preset precision bits number b, learnable scale parameter γ, and offset parameter β. This quantization enables us to encode the covariance with b-bit precision while storing the scaling factors in 32-bit floating point.

For the spectral attribute vector $\mathbf{c}_{\text{spec}}$, we utilize residual vector quantization (RVQ) [37] owing to their similarity among Gaussians. RVQ cascades M stages with a codebook of size B, which progressively refines quantization residuals over multiple stages. For simplicity, we denote $\mathbf{c}$ as $\mathbf{c}_{\text{spec}}$ for one Gaussian. The quantized output at the mth stage is calculated as

$$\hat{\mathbf{c}}_m = \textstyle\sum_{i=1}^{m} \mathbf{C}_i[t_i] \tag{10}$$

where $\mathbf{C}_i[j] \in \mathbb{R}^{B\times k}$ represents the jth row of the ith stage codebook $\mathbf{C}_i$. Here, t_i corresponds to the index of closest row vector in $\mathbf{C}_i$ from the residual spectral attribute $\mathbf{c} - \hat{\mathbf{c}}_{i-1}$, that is, $t_i = \arg\min_t \|\mathbf{C}_i[t] - (\mathbf{c} - \hat{\mathbf{c}}_{i-1})\|_2^2$. In addition, we set $\hat{\mathbf{c}}_0 = 0$.

To train the codebooks effectively, we use a commitment loss function by considering all the spectral attributes in Gaussians and all the stages

$$\mathcal{L}_c = \frac{1}{NM} \sum_{m=1}^{M} \sum_{n=1}^{N} \left\| \text{sg}\left(\mathbf{c}^{(n)} - \hat{\mathbf{c}}_{m-1}^{(n)}\right) - \mathbf{C}_m\left[t_m^{(n)}\right] \right\|_2^2 \tag{11}$$

where sg[·] is the stop-gradient operation. Consequently, we adopt a loss function consisting of both reconstruction loss and commitment loss when quantizing

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_c \tag{12}$$

where $\lambda > 0$ is a hyperparameter to trade off these two terms. Such involvement of RVQ mitigates the limitations of single-stage VQ by progressively refining quantization residuals, enabling higher fidelity under limited bitrates.

2) Entropy Coding: Once the attributes have been quantized, we further compress the discrete values using entropy coding. We adopt asymmetric numeral systems (ANSs) [38] to encode both the quantized covariance parameters and the indices of the RVQ codebooks, thereby exploiting statistical redundancy in the quantized values. Our whole pipeline is summarized in Algorithm 1.

V. Experiments

A. Data Description and Experimental Settings

1) Datasets: We evaluated our method on four widely used HSI datasets: Pavia University, Salinas, Jasper Ridge, and Urban.

1) *Pavia University*: Acquired over the University of Pavia, the dataset comprises 103 spectral bands with a spatial resolution of 610 × 340 pixels. We crop a 340 × 340 region from the lower portion for evaluation.
2) *Salinas*: A dataset acquired by the 224 spectral bands AVIRIS covering 512 × 217 pixels, and after removing water-absorption bands, it consists of 204 usable spectral bands. An area covering 180 × 180 pixels in the upper right corner of the image is selected for the experiment.
3) *Jasper Ridge*: This dataset contains 198 spectral bands ranging from 380 to 2500 nm after removing noisy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 LoR-SGS Optimization Algorithm

Require: HSI $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$,
NMF rank k, Size of Gaussians N,
Total iterations T, learning rate η,
Commitment weight λ
Ensure: Learned Gaussian parameters $\{\boldsymbol{\mu}^{(i)}, \mathbf{l}^{(i)}, \mathbf{c}_{\text{spec}}^{(i)}\}_{i=1}^{N}$
Step 1: Coefficient matrix estimation via NMF
Solve (6) to obtain $\mathbf{E}_0$
Step 2: Initialize Gaussians
for $i = 1, \ldots, N$ **do**
  Initialize center $\boldsymbol{\mu}^{(i)}$ uniformly over spatial domain
  Initialize Cholesky vector $\mathbf{l}^{(i)}$ as $[\delta, 0, \delta]$
  Initialize spectral vector $\mathbf{c}_{\text{spec}}^{(i)}$ uniformly over $[0, 1]$
end for
Step 3: Optimization Loop
for $t = 1 \rightarrow T$ **do**
  Implement quantization on Gaussians and compute quantization commitment loss $\mathcal{L}_c$ via (11)
  Decompress the Gaussians and compute $\hat{\mathcal{A}}$ via (7)
  Compute reconstruction loss $\mathcal{L}_{rec}$ via (9)
  Total loss $\mathcal{L} \leftarrow \mathcal{L}_{rec} + \lambda \mathcal{L}_c$
  Step 4: Parameter Updates
  for $i = 1, \ldots, N$ **do**
    $(\boldsymbol{\mu}^{(i)}, \mathbf{l}^{(i)}, \mathbf{c}_{\text{spec}}^{(i)}) \leftarrow (\boldsymbol{\mu}^{(i)}, \mathbf{l}^{(i)}, \mathbf{c}_{\text{spec}}^{(i)}) - \eta \nabla_{\boldsymbol{\Theta}} \mathcal{L}$
  end for
  if converged **then**
    break
  end if
end for
return $\{\boldsymbol{\mu}^{(i)}, \mathbf{l}^{(i)}, \mathbf{c}_{\text{spec}}^{(i)}\}_{i=1}^{N}$

bands with a spatial resolution of 100×100 pixels, capturing natural terrain features.

4) *Urban*: Consisting of 162 spectral bands ranging from 400 nm to 2500 nm after removing noisy bands, this dataset features urban scenes with a spatial resolution of 307×307 pixels.

2) Evaluation Metrics and Implementation Details: We evaluate our method using the following metrics.

1) *Bits-Per-Pixel-Per-Band*: Reflects the overall compression efficiency by accounting for the total bits used relative to the spatial and spectral dimensions, the bpppb is computed as

$$bpppb = \frac{\text{Bits for Gaussian} + 16\times k\times l}{h\times w\times l} \tag{13}$$

where "Bits for Gaussian" refers to the total number of bits required to encode all quantized Gaussian parameters, and the $16 \times k \times l$ term accounts for the bits used to encode the initial coefficient matrix $\mathbf{E}_0$ (at 16-bit precision per component). The size of the compressed HSI is primarily determined by the number of Gaussian ellipses, N. Adjusting N allows precise control over compression: higher N increases bpppb and reconstruction quality, while lower N enables stronger compression with reduced storage.

2) *Peak Signal-to-Noise Ratio*: Measures the fidelity of the reconstructed image relative to the ground truth

$$\text{PSNR} = \frac{1}{l}\sum_i 10\log_{10}\left(\frac{1}{\text{MSE}_i}\right) \tag{14}$$

where $\text{MSE}_i = 1/(hw)\|\mathcal{X}[i] - \hat{\mathcal{X}}[i]\|_F^2$, with $\mathcal{X}[i], \hat{\mathcal{X}}[i] \in \mathbb{R}^{h\times w}$ are the ith frontal slice of original and reconstructed HSIs.

3) *Multiscale Structural Similarity*: Evaluates the perceptual quality of the reconstructed image by measuring its structural similarity to the ground truth across multiple scales. At each scale, it computes luminance, contrast, and structure components using local statistics and combines them using weighted geometric means. The final multiscale structural similarity (MS-SSIM) score is calculated by aggregating these components across scales, where small constants and weighting factors are used to ensure stability and perceptual alignment. For visualization purposes, we plot $-10\log_{10}(1-\text{MS-SSIM})$ to magnify small perceptual differences.

4) *Spectral Angle Mapper* Measures the spectral distortion between the reconstructed and original hyperspectral pixels by computing the angle between spectral vectors

$$\text{SAM} = \frac{1}{hw}\arccos\left(\frac{\langle \mathbf{s}_j, \hat{\mathbf{s}}_j\rangle}{\|\mathbf{s}_j\|_2 \cdot \|\hat{\mathbf{s}}_j\|_2 + \epsilon}\right) \tag{15}$$

where $\mathbf{s}_j$ and $\hat{\mathbf{s}}_j$ denote the jth spectral vectors of the original and reconstructed HSIs, respectively. Here, ϵ is a small constant to avoid division by zero.

For implementation, we employ the Adam optimizer with an initial learning rate of 5×10^{-3}. The total training iterations are set to 8000, 7000, 7000, and 7000 iterations on the Jasper Ridge, Salinas, Pavia University, and Urban datasets, respectively. The learning rate is decayed by a factor of 0.5 every 3000 steps.

3) Compared Methods: We evaluated our LoR-SGS against several compression methods of different types, including both conventional codecs (JPEG [6], JPEG2000 [7], PCA-DCT [8]) and state-of-the-art INR-based methods (INR by Rezasoltani and Qureshi [22] and HINER [23]). To ensure a fair comparison while accounting for computational requirements, the conventional methods were evaluated on a Windows 11 system with 16GB RAM, while INR, HINER, and LoR-SGS were assessed using an NVIDIA L40 GPU with 48GB memory.

4) Hyperparameter Selection: To accommodate the distinctive spatial and spectral traits within individual datasets, hyperparameters are carefully tuned to ensure high reconstruction fidelity. In our experiments, we employed a two-step methodology to optimize our model's performance: First, we identified the optimal low-rank value k by testing several candidates. For each k, we scaled the number of Gaussians (N) to meet a specific bitrate target and selected the k that yielded the highest PSNR. Subsequently, with this optimal k fixed, we generated the final rate-distortion curves. This was achieved by simultaneously fine-tuning both the number of Gaussians (N) and the codebook size (B), enabling precise control over the bitrate and allowing us to maximize the PSNR across the entire bitrate spectrum.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

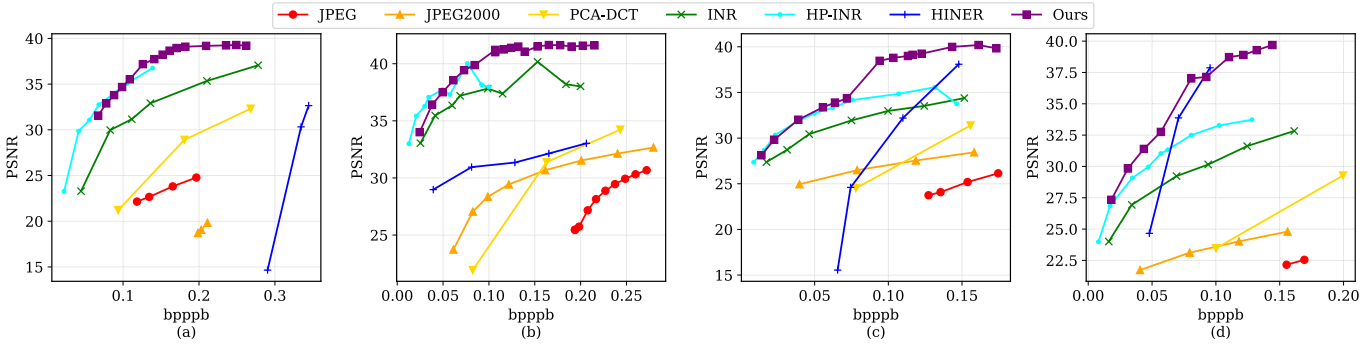


Fig. 2. Compression results: PSNR values achieved at various bpppb for different methods on (a) Jasper ridge, (b) Salinas, (c) Pavia university, and (d) Urban datasets. Note: "HP-INR" denotes HP INR using 16-bit parameter storage for fair comparison.

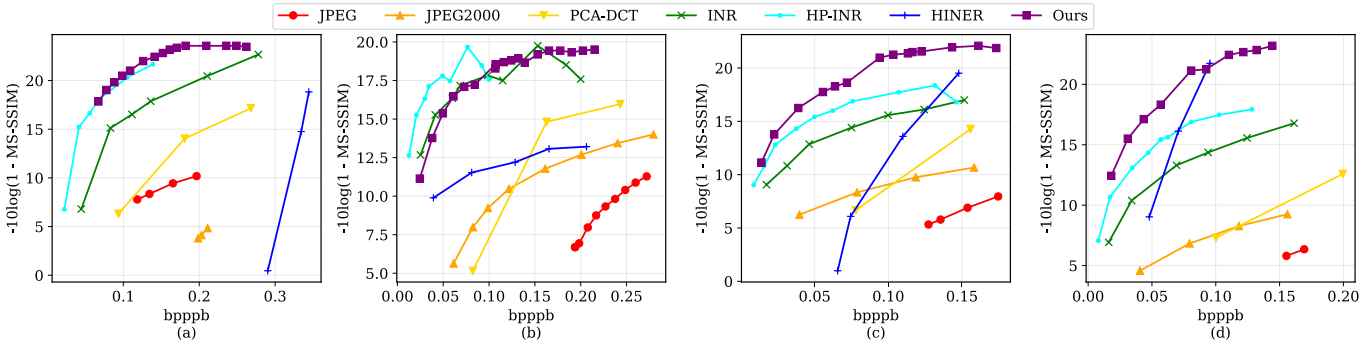


Fig. 3. Compression results: MS-SSIM values in log-scale achieved at various bpppb for different methods on (a) Jasper ridge, (b) Salinas, (c) Pavia university, and (d) Urban datasets. Note: "HP-INR" denotes HP INR using 16-bit parameter storage for fair comparison.

TABLE I
HYPERPARAMETERS FOR EACH DATASET

	PaviaU	Salinas	Jasper Ridge	Urban
Rank k	12	12	10	12
Size N at high-bpppb	23000	13000	2750	29000
Size N at low-bpppb	10000	5000	1000	14500
λ	0.6	0.1	0.05	0.65
Codebook size B	64	72	72	64

The values of hyperparameters are detailed in Table I. We also report the exact number of Gaussians N used to achieve each target bpppb. Common settings across all datasets include a $b = 8$ in UQ, a quantization stage count of $M = 2$, and eight iterations for the K-means algorithm used in RVQ [37].

B. Main Experimental Results

1) Quantitative Comparison:

a) Comparison on compression performance: Table II presents quantitative comparison results on four HSI datasets at matched bitrates (0.1 and 0.05/0.06 bpppb for deep methods). At equivalent compression levels, our proposed LoR-SGS consistently achieves superior performance across all metrics. It secures the highest PSNR and MS-SSIM values, coupled with significantly lower spectral angle mapper (SAM) scores, indicating highly accurate spatial and spectral reconstruction. Notably, compared to the leading INR-based method [22] at these bitrates, LoR-SGS achieves significant quality enhancements, with PSNR improvements of up to 4 dB, MS-SSIM gains of 0.01, and SAM reductions of 0.02 across all four datasets. These significant gains are primarily attributed to the dynamic and adaptive representational capacity of our SGS.

Figs. 2–4 illustrate the PSNR, log-scale MS-SSIM, and SAM values across various bitrates for all methods. Our LoR-SGS consistently outperforms HP-INR, JPEG2000, and other baselines across the entire bitrate range. For example, Fig. 2(c) shows that at 0.1 bpppb, LoR-SGS achieves 38.95 dB on Pavia University, significantly surpassing HP-INR (35 dB). In Fig. 3, LoR-SGS maintains a log-scale MS-SSIM above 15 at 0.05 bpppb across all datasets, outperforming HP-INR, which falls below 15 on most datasets. SAM results (Fig. 4) show LoR-SGS achieves the lowest spectral error across all bitrates, highlighting its superior spectral fidelity and robust low-rank modeling.

Fig. 5 provides a detailed breakdown of PSNR across all spectral bands for each method. In every scene, our method remains above 30 dB concerning PSNR. In contrast, traditional codecs like JPEG, JPEG2000, and PCA-DCT show sharp PSNR drops of 5–10 dB at certain wavelengths, indicating severe distortion. While INR performs better than traditional methods, its PSNR remains below LoR-SGS, highlighting its limited capacity for adaptive reconstruction. Overall, LoR-SGS's smooth and elevated PSNR curve across all wavelengths demonstrates its robustness in preserving spectral details.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPRESSION PERFORMANCE COMPARISON ON FOUR HSI DATASETS, THE BEST RESULTS ARE BOLD-FACED. NOTE THAT HINER RESULTS FOR JASPER RIDGE AT THESE SPECIFIC BITRATES WERE NOT AVAILABLE

Method	Pavia University					Salinas				
	bpppb	Size(KB)	PSNR	MS-SSIM	SAM	bpppb	Size(KB)	PSNR	MS-SSIM	SAM
JPEG	0.1	145	23.76	0.7019	0.2454	0.1	81	25.46	0.7858	0.1112
JPEG2000	0.1	145	27.11	0.8790	0.1398	0.1	81	28.36	0.8804	0.0541
PCA-DCT	0.1	145	24.70	0.7999	0.2270	0.1	81	21.94	0.6955	0.0981
INR	0.1	145	33.15	0.9760	0.0802	0.1	81	37.38	0.9822	0.0257
HP-INR	0.06	87	32.83	0.9747	0.0812	0.05	40	37.30	0.9821	0.0258
HINER	0.1	145	32.20	0.9532	0.2582	0.1	81	31.35	0.9398	0.0609
	0.06	87	15.55	0.2040	0.6934	0.05	40	28.98	0.8975	0.0597
Ours	0.1	145	**38.95**	**0.9926**	**0.0564**	0.1	81	**41.41**	**0.9916**	**0.0135**
	0.06	87	**35.76**	**0.9881**	**0.0697**	0.05	40	**37.83**	**0.9869**	**0.0170**

Method	Jasper Ridge					Urban				
	bpppb	Size(KB)	PSNR	MS-SSIM	SAM	bpppb	Size(KB)	PSNR	MS-SSIM	SAM
JPEG	0.1	24	21.15	0.8267	0.2177	0.1	186	22.15	0.7358	0.2535
JPEG2000	0.1	24	18.71	0.5817	0.2800	0.1	186	23.70	0.8314	0.1616
PCA-DCT	0.1	24	28.91	0.9605	0.1018	0.1	186	23.47	0.8141	0.2758
INR	0.1	24	31.34	0.9818	0.0706	0.1	186	31.20	0.9733	0.0862
HP-INR	0.06	15	**31.25**	**0.9816**	0.0709	0.05	93	30.79	0.9721	0.0874
HINER	0.1/0.06	24/15	–	–	–	0.05	93	24.66	0.8751	1.5708
Ours	0.1	24	**37.19**	**0.9936**	**0.0380**	0.1	186	**38.22**	**0.9940**	**0.0468**
	0.06	15	30.79	0.9804	**0.0598**	0.05	93	**34.24**	**0.9883**	**0.0652**

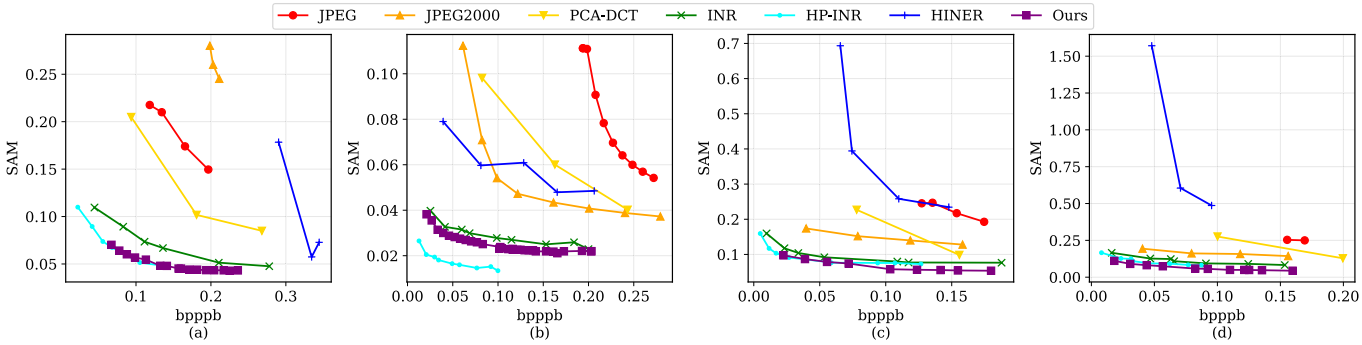


Fig. 4. Compression results: SAM values achieved at various bpppb for different methods on (a) Jasper ridge, (b) Salinas, (c) Pavia university, and (d) Urban datasets. Note: "HP-INR" denotes HP INR using 16-bit parameter storage for fair comparison.

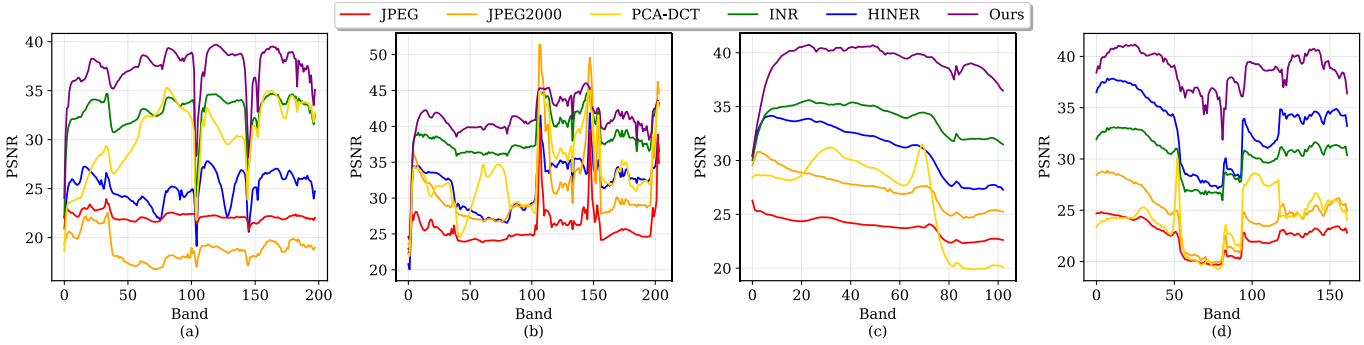


Fig. 5. Compression results: PSNR values in different spectral bands achieved at ~0.1 bpppb for different methods (~0.2 bpppb for INR on jasper ridge) on (a) Jasper ridge, (b) Salinas, (c) Pavia university, and (d) Urban datasets.

b) Encoding/Decoding efficiency and model complexity: Table III summarizes encoding/decoding times, model parameters, and GPU memory usage for all methods. LoR-SGS achieves competitive encoding times, significantly faster than HINER and comparable to INR. Traditional codecs like JPEG perform fast but with poor compression and reconstruction quality. Importantly, LoR-SGS excels in decoding, taking under 0.02 s, matching INR and HINER speeds while being 10–100 times faster than JPEG2000 and PCA-DCT, making it ideal for real-time applications. In terms of model complexity, LoR-SGS uses 15–476 K parameters, occasionally exceeding INR/HP-INR due to its flexible Gaussian representation, but significantly reduces GPU memory usage compared to INR/HP-INR and HINER. This advantageous trade-off

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

TABLE III

TIME OF COMPRESSION/DECOMPRESSION AND NUMBERS OF MODEL PARAMETERS/GPU MEMORY CONSUMPTION DURING TRAINING ACROSS FOUR DATASETS. THE BEST RESULTS ARE BOLD-FACED. NOTE THAT HINER RESULTS FOR JASPER RIDGE AT THESE SPECIFIC BITRATES WERE NOT AVAILABLE

Pavia University							
	Method	bpppb	Comp. time(s)	Decomp. time(s)	PSNR	Params(K)	GPU Mem(MiB)
Conventional	JPEG	0.1	**0.0206**	0.0139	23.76	–	–
	JPEG2000	0.1	1.1105	0.0622	27.11	–	–
	PCA-DCT	0.1	0.0780	0.0319	24.70	–	–
Learning-based	INR	0.1	149.94	**0.0007**	33.15	**43.7**	1986
	HP-INR	0.06	149.94	0.1159	32.84	**43.7**	1986
	HINER	0.1	163.26	0.0015	32.20	580	1048
		0.06	163.34	**0.0015**	15.55	280	1058
	Ours	0.1	47.53	0.0167	**38.95**	391	**876**
		0.06	**47.87**	0.0087	**35.76**	170	**878**
Salinas							
	Method	bpppb	Comp. time(s)	Decomp. time(s)	PSNR	Params(K)	GPU Mem(MiB)
Conventional	JPEG	0.1	**0.0134**	0.0093	25.46	–	–
	JPEG2000	0.1	0.6192	0.0327	28.36	–	–
	PCA-DCT	0.1	0.0400	0.0165	21.94	–	–
Learning-based	INR	0.1	58.58	**0.0012**	37.38	**23.7**	862
	HP-INR	0.05	58.58	0.1123	37.30	**23.7**	862
	HINER	0.1	127.17	0.0015	31.35	480	788
		0.05	125.28	**0.0018**	28.98	180	784
	Ours	0.1	45.84	0.0102	**41.41**	221	**706**
		0.05	**45.10**	0.0054	**37.83**	85	**696**
Jasper Ridge							
	Method	bpppb	Comp. time(s)	Decomp. time(s)	PSNR	Params(K)	GPU Mem(MiB)
Conventional	JPEG	0.1	**0.0058**	0.0043	21.15	–	–
	JPEG2000	0.1	0.2930	0.0150	18.71	–	–
	PCA-DCT	0.1	0.0358	0.0057	28.91	–	–
Learning-based	INR	0.1	52.53	**0.0007**	31.34	**8.42**	620
	HP-INR	0.06	52.53	0.1167	**31.25**	**8.42**	620
	HINER	0.1/0.06	–	–	–	–	–
	Ours	0.1	52.17	0.0044	**37.19**	41	**600**
		0.06	**50.22**	**0.0031**	30.79	15	**598**
Urban							
	Method	bpppb	Comp. time(s)	Decomp. time(s)	PSNR	Params(K)	GPU Mem(MiB)
Conventional	JPEG	0.1	**0.0277**	0.0196	22.15	–	–
	JPEG2000	0.1	1.5929	0.0775	23.70	–	–
	PCA-DCT	0.1	0.0895	0.0447	23.47	–	–
Learning-based	INR	0.1	60.06	**0.0010**	31.20	**55.1**	1660
	HP-INR	0.05	60.06	0.1058	30.79	**55.1**	1660
	HINER	0.1	192.48	0.0015	37.87	970	1316
		0.05	189.01	**0.0015**	24.66	480	1318
	Ours	0.1	49.61	0.0196	**38.22**	476	**980**
		0.05	**49.61**	0.0099	**34.24**	204	**914**

between efficiency, resource consumption, and high-quality reconstruction positions LoR-SGS.

2) Qualitative Evaluation: To complement our quantitative analysis, we conducted a qualitative evaluation by visualizing the reconstructed spatial structures of multiple datasets.

a) Pseudo color images and error maps of MSE: Figs. 6–9 provide visual comparisons, presenting pseudo-color images alongside their corresponding per-pixel absolute error heatmaps for each dataset at approximately 0.1 bpppb. These visual representations demonstrate LoR-SGS's remarkable performance in reconstructing homogeneous areas, such as water bodies, large agricultural fields, or broad asphalt surfaces. These regions remain virtually indistinguishable from the original HSI, a testament to the stability of our quantization scheme. This stability allows the Gaussian primitives to represent low-variance signals with minimal distortion, thereby effectively preserving low-frequency content by adeptly handling small spectral variations.

C. Ablation Studies

To thoroughly understand and quantify the individual contributions of LoR-SGS's core components, we conducted a series of comprehensive ablation studies. Specifically, our analysis focused on evaluating the optimal choice of rank in the matrix decomposition, the RVQ codebook size, various estimation methods for the coefficient matrix $\mathbf{E}_0$, as well as the necessity and efficacy of each quantization component.

1) Rank of Matrix Decomposition: We systematically varied the matrix rank $k \in \{4, 6, 8, 10, 12, 16\}$ and analyzed its effect on PSNR across the four datasets, as depicted in Fig. 10. For instance, on the Jasper Ridge dataset, PSNR peaks at $k = 12$. This suggests that using a rank approximately twice that of the often-estimated intrinsic spectral subspace (typically around 6 in related studies) yields optimal reconstruction error for this dataset. Across most datasets, performance tends to saturate beyond a certain optimal k value identified in our experimental setting, with further gains diminishing for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

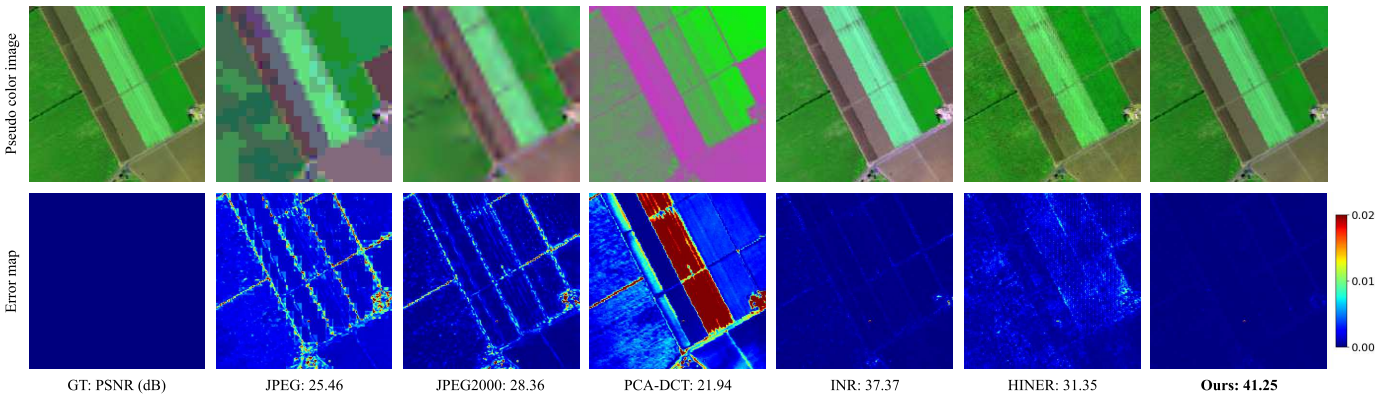


Fig. 6. Salinas dataset: pseudo-color (bands 11/91/181) versus MSE error map at ~0.1 bpppb for our LoR-SGS and baselines.

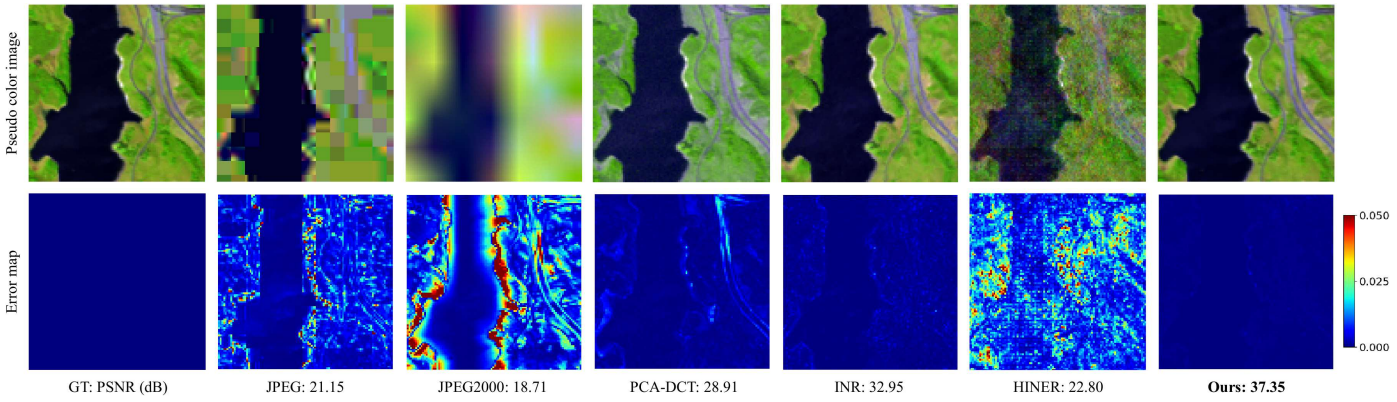


Fig. 7. Jasper ridge dataset: pseudo-color (bands 30/80/130) versus MSE error map at ~0.1 bpppb for our LoR-SGS and baselines (HINER was implemented at ~0.2 bpppb).

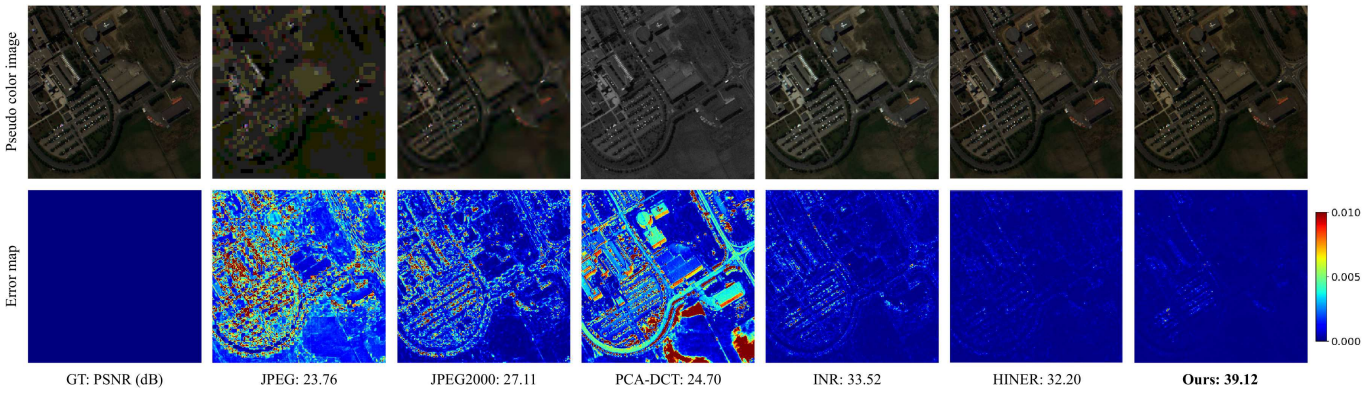


Fig. 8. Pavia university dataset: pseudo-color (bands 11/31/51) versus MSE error map at ~0.1 bpppb for our LoR-SGS and baselines.

larger k values. This observed behavior directly reflects a well-known trade-off in NMF: a rank that is too low can lead to underfitting of the inherent spectral complexity, whereas a rank that is excessively high unnecessarily increases the number of model parameters and computational burden without providing a significant PSNR improvement.

2) Type of Estimation Method on Coefficient Matrix: On the Salinas dataset, we evaluated four methods for estimating the coefficient matrix $\mathbf{E}_0$: NMF, VCA, PCA, and SVD. As shown in Table IV, NMF delivers the best performance with a PSNR of 41.25 dB, MS-SSIM of 0.9921, and the lowest SAM of 0.0231. These results highlight the importance of interpretable spectral bases, such as NMF

TABLE IV

COMPARISON OF COMPRESSION PERFORMANCE WITH DIFFERENT COEFFICIENT ESTIMATION METHODS ON SALINAS DATASET

Method	PSNR	MS-SSIM	SAM
SVD	11.78	0.4074	1.4315
PCA	29.16	0.9818	0.1503
VCA	36.83	0.9818	0.0431
NMF	**41.25**	**0.9921**	**0.0231**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

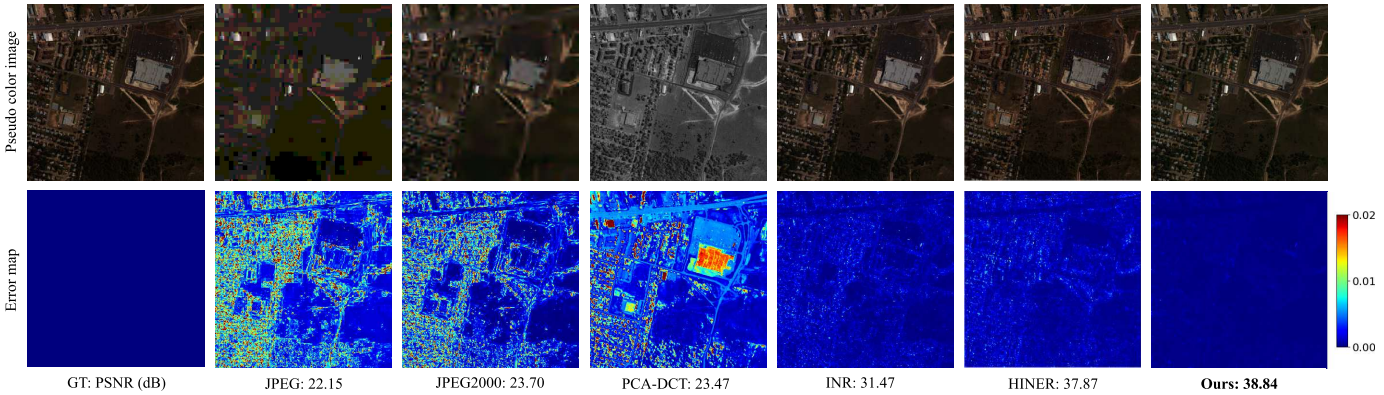


Fig. 9. Urban dataset: pseudo-color (bands 11/31/51) versus MSE error map at ~0.1 bpppb for our LoR-SGS and baselines.

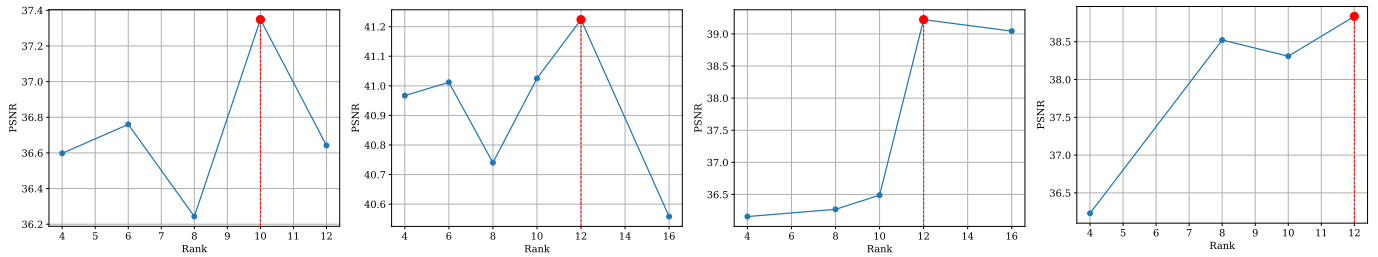


Fig. 10. Compression performance with different rank values on (a) Jasper ridge, (b) Salinas, (c) Pavia university, and (d) Urban datasets.

TABLE V

COMPARISON OF COMPRESSION PERFORMANCE ACROSS CODEBOOK SIZES B FOR RVQ ON C

B	72	64	48	32	24	16	8
bpppb	0.1174	0.1132	0.1109	0.1073	0.1051	0.1081	**0.0982**
PSNR	**39.38**	38.52	38.26	37.92	37.51	37.70	35.38

TABLE VI

ABLATION STUDY ON QUANTIZATION STRATEGIES (JASPER RIDGE DATASET)

	N	bpppb	PSNR	MS-SSIM	SAM
16-bit + UQ	600	0.1	31.45	0.9844	0.0542
16-bit + UQ	2700	0.5	40.84	0.9970	0.0314
16-bit + RVQ		0.25	37.92	0.9942	0.0489
UQ + RVQ		0.2	37.10	0.9936	0.0503
16-bit + UQ + RVQ		0.1	**36.75**	**0.9934**	**0.0509**
16-bit + UQ + RVQ + w/o ANS		0.3	36.86	0.9934	0.0507

and VCA, for high-fidelity HSI reconstruction, whereas traditional methods like SVD and PCA lack the discriminative power needed for compact and accurate Gaussian modeling.

3) Codebook Size for RVQ [37]: Table V presents the impact of varying the RVQ codebook size B on compression performance, fixing the number of Gaussians N as 29 000. When B decreases from 72 to 8, the PSNR on the Urban dataset (at $\approx$ 0.11 bpppb) degrades from 39.38 to 35.38 dB, demonstrating a clear positive correlation between codebook capacity and reconstruction fidelity. A notable observation is the steepest PSNR drop when B falls below 16, which indicates a critical threshold below which residual quantization struggles to adequately capture complex color residuals. Interestingly, reducing B from 72 to 48 incurs a comparatively minor PSNR loss of approximately 1.12 dB, suggesting diminishing returns on very large codebooks for PSNR. This ablation study highlights that a moderate codebook size, such as $B = 48$, can achieve a significant portion of the available PSNR gain while incurring lower metadata overhead. Conversely, using very small codebooks ($B \leq 16$) sharply compromises reconstruction quality. This behavior reflects a critical hyperparameter trade-off in RVQ.

4) Quantization Strategy: We evaluated the impact of our quantization scheme components on the Jasper Ridge dataset, as shown in Table VI. Our full scheme (16-bit precision + UQ + RVQ) serves as the baseline. Here, we simplified specific elements while keeping the Gaussian count constant ($N = 2700$) for fair comparison. When 16-bit precision was removed (UQ + RVQ), bpppb increased to 0.2, but PSNR remained high at 37.10 dB, showing 16-bit precision is critical for file size reduction. Removing UQ for covariance (16-bit precision + RVQ) raised bpppb to 0.25, with PSNR improving slightly to 37.92 dB, suggesting UQ plays a less significant role. Without RVQ (16-bit precision + UQ), PSNR increased to 40.84 dB, but bpppb rose sharply to 0.5, highlighting that unquantized spectral attributes achieve higher quality but are inefficient for compression. Moreover, when RVQ was removed and the Gaussian count adjusted to match the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

baseline bpppb (0.1), PSNR dropped significantly to 31.45 dB, demonstrating a 5.3 dB loss and reduced MS-SSIM. Finally, removing the ANS entropy coding resulted in a bpppb of 0.3128 (PSNR 36.74 dB), confirming that entropy coding is crucial for achieving high compression ratios. These results clearly show that our full quantization scheme, especially RVQ and ANS, is essential for achieving high fidelity at low bitrates. Simpler approaches, akin to those used in some INR-based methods [22], [23], fail to balance quality and compression efficiency effectively.

VI. Conclusion

We have presented LoR-SGS, a hybrid HSI compression pipeline that first decouples spectral and spatial redundancies via interpretable NMF factorization into endmember and abundance matrices, then adaptively captures fine spatial detail by modeling abundances with learned 2-D Gaussians, and finally compresses all parameters through RVQ and ANS entropy coding. Regarding the limitations of this work, it is worth mentioning that our current implementation fully relies on a GPU for efficiency, making it unsuitable for resource-constrained edge devices. In addition, the decoding pipeline, which involves reconstructing attributes from quantized data, presents a trade-off in speed. Furthermore, In future studies, we will explore optimizing the decoding process and adapting our method for CPU-based environments. We also believe that the principles of our SGS can be applied to other tasks, including hyperspectral super-resolution and hyperspectral and multispectral image fusion.

References

[1] A. Bhargava, A. Sachdeva, K. Sharma, M. H. Alsharif, P. Uthansakul, and M. Uthansakul, "Hyperspectral imaging and its applications: A review," *Heliyon*, vol. 10, no. 12, Jun. 2024, Art. no. e33208.

[2] B. G. Ram, P. Oduor, C. Igathinathane, K. Howatt, and X. Sun, "A systematic review of hyperspectral imaging in precision agriculture: Analysis of its current state and future prospects," *Comput. Electron. Agricult.*, vol. 222, Jul. 2024, Art. no. 109037.

[3] C.-L. Lai et al., "Advancing hyperspectral imaging and machine learning tools toward clinical adoption in tissue diagnostics: A comprehensive review," *APL Bioeng.*, vol. 8, no. 4, Dec. 2024, Art. no. 041504.

[4] A. Altamimi and B. Youssef, "A systematic review of hardware-accelerated compression of remotely sensed hyperspectral images," *Sensors*, vol. 22, no. 1, p. 263, 2021.

[5] Y. Dua, V. Kumar, and R. S. Singh, "Comprehensive review of hyperspectral image compression algorithms," *Opt. Eng.*, vol. 59, no. 9, Sep. 2020, Art. no. 090902.

[6] W. F. Good, G. S. Maitz, and D. Gur, "Joint photographic experts group (JPEG) compatible data compression of mammograms," *J. Digit. Imag.*, vol. 7, no. 3, pp. 123–132, Aug. 1994.

[7] Q. Du and J. E. Fowler, "Hyperspectral image compression using JPEG2000 and principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 2, pp. 201–205, Apr. 2007.

[8] Y. Nian, Y. Liu, and Z. Ye, "Pairwise KLT-based compression for multispectral images," *Sens. Imag.*, vol. 17, no. 1, pp. 1–15, Dec. 2016.

[9] T. Qiao, J. Ren, M. Sun, J. Zheng, and S. Marshall, "Effective compression of hyperspectral imagery using an improved 3D DCT approach for land-cover analysis in remote-sensing applications," *Int. J. Remote Sens.*, vol. 35, no. 20, pp. 7316–7337, Oct. 2014.

[10] R. Dusselaar, M. Paul, and T. Bossomaier, "Hyperspectral image coding using spectral prediction modelling in HEVC coding framework," in *Proc. Int. Conf. Image Vis. Comput. New Zealand (IVCNZ)*, Nov. 2015, pp. 1–6.

[11] L. Santos, S. Lopez, G. M. Callico, J. F. Lopez, and R. Sarmiento, "Performance evaluation of the H.264/AVC video coding standard for lossy hyperspectral image compression," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 451–461, Apr. 2012.

[12] D. Valencia and A. Plaza, "FPGA-based hyperspectral data compression using spectral unmixing and the pixel purity index algorithm," in *Proc. 6th Int. Conf. Comput. Sci. (ICCS)*, May 2006, pp. 888–891.

[13] A. Plaza, "Towards real-time compression of hyperspectral images using Virtex-II FPGAs," in *Proc. 13th Int. Euro-Par Parallel Process. Conf.*, Aug. 2007, pp. 248–257.

[14] B. Du, M. Zhang, L. Zhang, R. Hu, and D. Tao, "PLTD: Patch-based low-rank tensor decomposition for hyperspectral images," *IEEE Trans. Multimedia*, vol. 19, no. 1, pp. 67–79, Jan. 2017.

[15] Y. Chong, W. Zheng, H. Li, and S. Pan, "Block-sparse tensor based spatial–spectral joint compression of hyperspectral images," in *Proc. 14th Int. Conf. Intell. Comput. Methodol.*, vol. 10969. Cham, Switzerland: Springer, Aug. 2018, pp. 260–265.

[16] S. Das, "Hyperspectral image, video compression using sparse tucker tensor decomposition," *IET Image Process.*, vol. 15, no. 4, pp. 964–973, Mar. 2021.

[17] D. Valsesia, T. Bianchi, and E. Magli, "Onboard deep lossless and near-lossless predictive coding of hyperspectral images with line-based attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532714.

[18] M. H. P. Fuchs, A. P. Byju, A. Walda, B. Rasti, and B. Demir, "Generative adversarial networks for spatio-spectral compression of hyperspectral images," in *Proc. 14th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.*, Dec. 2024, pp. 1–5.

[19] J. Kuester, W. Gross, S. Schreiner, W. Middelmann, and M. Heizmann, "Adaptive two-stage multisensor convolutional autoencoder model for lossy compression of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530022.

[20] Y. Guo, Y. Tao, Y. Chong, S. Pan, and M. Liu, "Edge-guided hyperspectral image compression with interactive dual attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500817.

[21] Y. Guo, Y. Chong, and S. Pan, "Hyperspectral image compression via cross-channel contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513918.

[22] S. Rezasoltani and F. Z. Qureshi, "Hyperspectral image compression using implicit neural representations," in *Proc. 20th Conf. Robots Vis. (CRV)*, Jun. 2023, pp. 248–255.

[23] J. Shi, M. Jiang, M. Lu, T. Chen, X. Cao, and Z. Ma, "HINER: Neural representation for hyperspectral image," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 9837–9846.

[24] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 1–14, Jul. 2023.

[25] X. Zhang et al., "GaussianImage: 1000 FPS image representation and compression by 2D Gaussian splatting," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 327–345.

[26] Y.-X. Wang and Y.-J. Zhang, "Nonnegative matrix factorization: A comprehensive review," *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 6, pp. 1336–1353, Jun. 2013.

[27] T. Wu et al., "Recent advances in 3D Gaussian splatting," *Comput. Vis. Media*, vol. 10, no. 4, pp. 613–642, Aug. 2024.

[28] J. C. Lee, D. Rho, X. Sun, J. H. Ko, and E. Park, "Compact 3D Gaussian representation for radiance field," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 21719–21728.

[29] Y. Chen et al., "GaussianEditor: Swift and controllable 3D editing with Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 21476–21485.

[30] J. Waczyńska, T. Szczepanik, P. Borycki, S. Tadeja, T. Bohne, and P. Spurek, "MiraGe: Editable 2D images using Gaussian splatting," 2024, *arXiv:2410.01521*.

[31] L. Zhu et al., "Large images are Gaussians: High-quality large image representation with levels of 2D Gaussian splatting," 2025, *arXiv:2502.09039*.

[32] M. Liu et al., "D2GV: Deformable 2D Gaussian splatting for video representation in 400FPS," 2025, *arXiv:2503.05600*.

[33] I. Lee, Y. Choi, and J. Lee, "GaussianVideo: Efficient video representation and compression by Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2025, pp. 4471–4480.

[34] M. Zwicker, H. Pfister, J. van Baar, and M. Gross, "EWA volume splatting," in *Proc. Visualizat.*, Sep. 2001, pp. 29–538.

[35] C.-J. Hsieh and I. S. Dhillon, "Fast coordinate descent methods with variable selection for non-negative matrix factorization," in *Proc. 17th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2011, pp. 1064–1072.

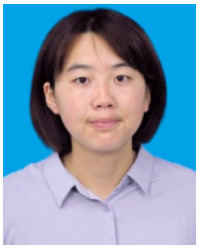
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.

[36] Y. Bhalgat, J. Lee, M. Nagel, T. Blankevoort, and N. Kwak, "LSQ+: Improving low-bit quantization through learnable offsets and better initialization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 2978–2985.
[37] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "SoundStream: An end-to-end neural audio codec," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 30, pp. 495–507, 2022.
[38] J. Duda, "Asymmetric numeral systems," 2009, *arXiv:0902.0271*.



Tianyu Li is currently pursuing the B.Sc. degree with the Department of Statistics and Data Science, Southern University of Science and Technology (SUSTech), Shenzhen, China.

He is also a Visiting Student with the University of California, Berkeley, CA, USA. His research interests include machine learning, computer vision, and hyperspectral image processing.



Ting Wang received the M.Sc. degree from the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an, China, in 2021. She is currently pursuing the Ph.D. degree with the Department of Statistics and Data Science, Southern University of Science and Technology (SUSTech), Shenzhen, China.

Her research interests include hyperspectral image processing and remote sensing data fusion.



Xile Zhao (Senior Member, IEEE) received the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2009 and 2012, respectively.

He worked as a Post-Doctoral Researcher with Prof. M. Ng with Hong Kong Baptist University, Hong Kong, from 2013 to 2014. He worked as a Visiting Scholar with Prof. J. Bioucas Dias with the University of Lisbon, Lisbon, Portugal, from 2016 to 2017. He is currently a Professor with the School of Mathematical Sciences, UESTC. His research interests include image processing, machine learning, and scientific computing. More information can be found on his homepage https://zhaoxile.github.io/



Chao Wang received the Ph.D. degree in mathematics from The Chinese University of Hong Kong, Hong Kong, in 2017.

He is currently an Assistant Professor with the Southern University of Science and Technology (SUSTech), Shenzhen, China. His research interests include scientific computing, compressed sensing, interdisciplinary mathematical modeling, convex and nonconvex optimization, medical imaging, machine learning, and numerical linear algebra.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:13:22 UTC from IEEE Xplore. Restrictions apply.